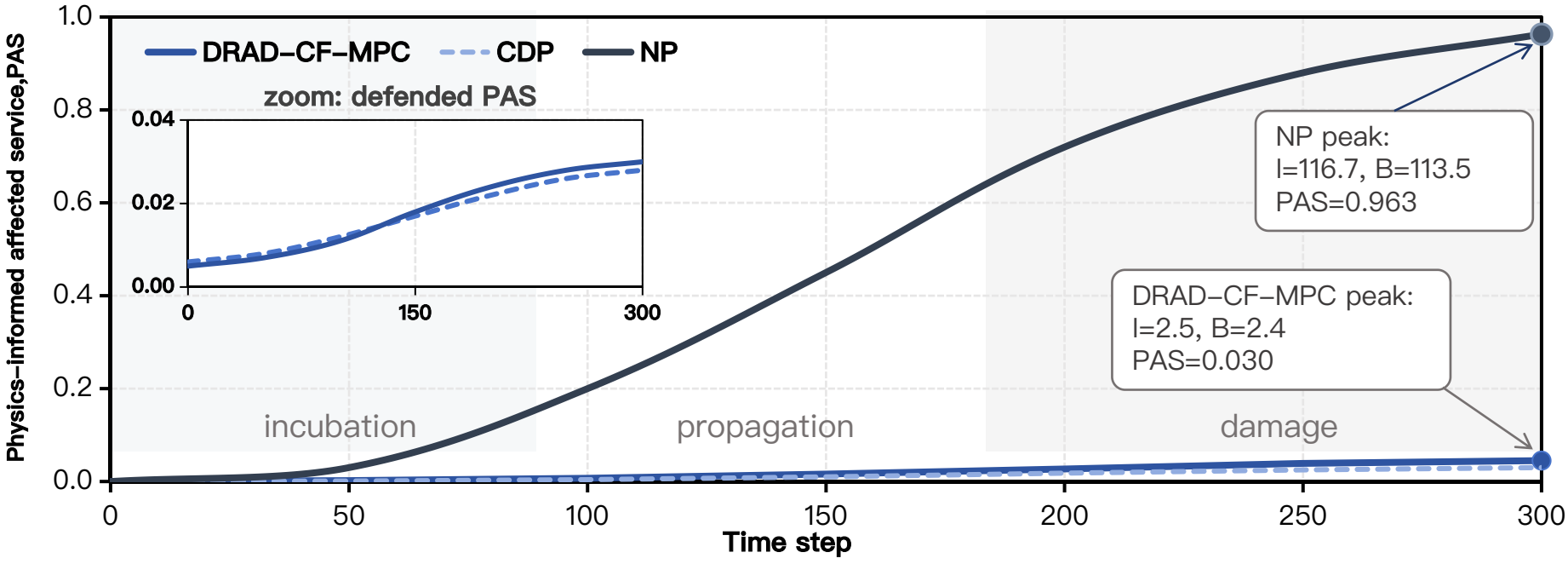
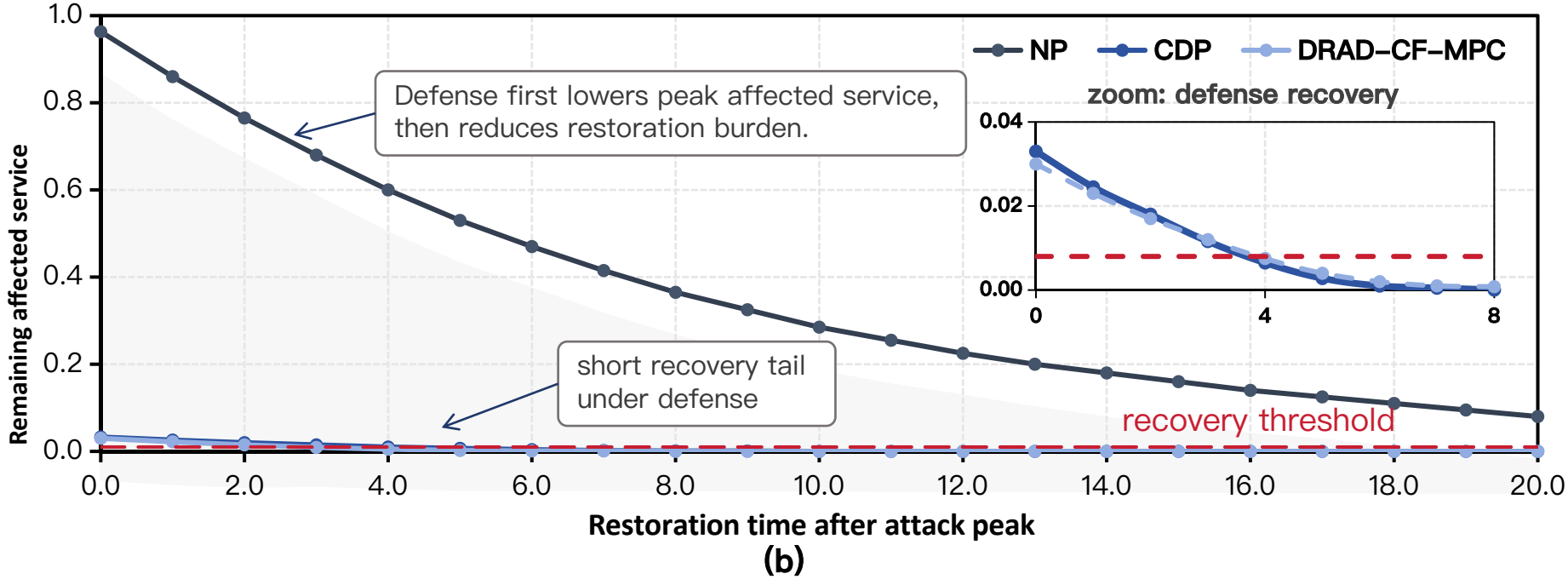

DRAD–CF–MPC
CDP
NP
zoom: defended PAS
0.04
0.02
0.00
0
150
300
NP peak:
I=116.7, B=113.5
PAS=0.963
DRAD–CF–MPC peak:
I=2.5, B=2.4
PAS=0.030
incubation
propagation
damage
Physics–informed affected service,PAS
1.0
0.8
0.6
0.4
0.2
0.0
0
50
100
150
200
250
300
Time step

(a)


NP
CDP
DRAD–CF–MPC
zoom: defense recovery
0.04
0.02
0.00
0
4
8
Defense first lowers peak affected service,
then reduces restoration burden.
short recovery tail
under defense
recovery threshold
Remaining affected service
1.0
0.8
0.6
0.4
0.2
0.0
0.0
2.0
4.0
6.0
8.0
10.0
12.0
14.0
16.0
18.0
20.0
Restoration time after attack peak

(b)